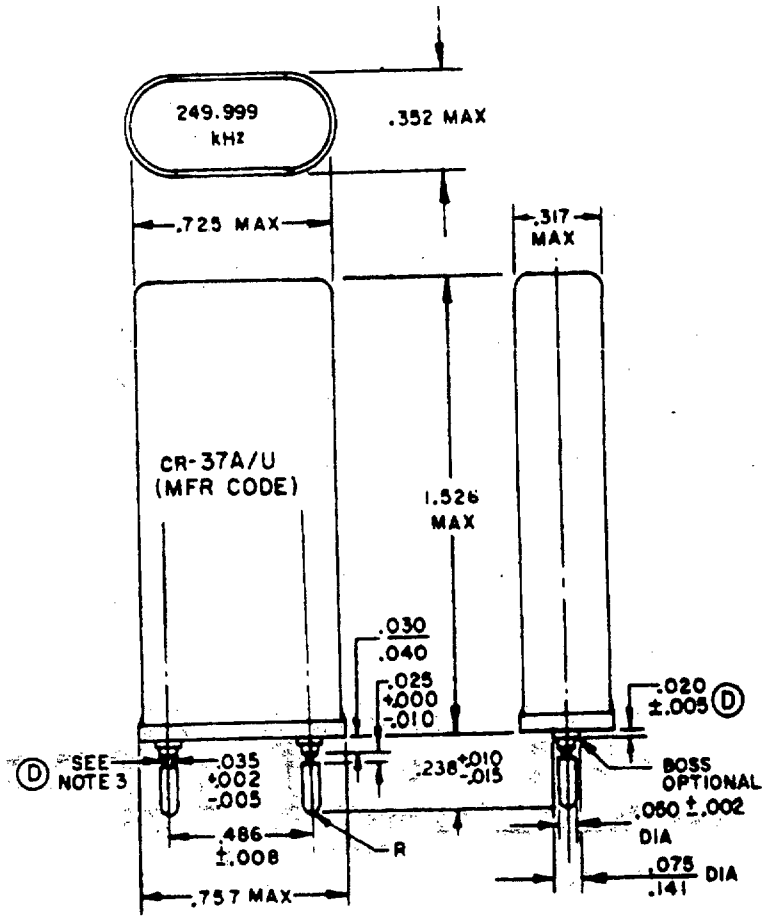
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-37A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 90 to 250 kHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.352	8.94
.005	.13	.025	.64	.075	1.91	.486	12.34
.008	.20	.030	.76	.141	3.58	.725	18.42
.010	.25	.035	.89	.238	6.05	.757	19.23
.015	.38	.040	1.02	.317	8.05	1.526	38.76

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
Ⓓ 3. The pin undercut may be omitted.

Ⓓ denotes change.

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FSC 5955

TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-13/U (MIL-H-10056/4)
Dimensions, marking, and configuration	See figure
Frequency range	90 to 250 kHz, inclusive
Capacitance, shunt	See table III
Frequency tolerance (operating temperature range)	±200 PPM
Equivalent resistance:	kHz — Ohms, maximum 90 to 170 — 5,000 170+ to 250 — 5,500
Mode of oscillation	Fundamental
Antiresonance, load capacitance	20.0 ± 0.5 pF
Operating temperature range	-40° to 70°C, inclusive
Test set, calibration values, and rated drive level	See table II
Shock (specified pulse):	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Vibration: 1/	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Temperature run:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Bond strength:	kHz — Grams, maximum 90 to 170 — 800 170+ to 250 — 700

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (total excursion 0.030 inch).

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	Ohms	mA	Volts	mW
TS-710/TSM	90 to 170	3500	---	2.65	2.0 ±0.4
	170+ to 250	3900	---	2.79	2.0 ±0.4

TABLE III. Capacitance, shunt.

Frequency range	Capacitance 1/
kHz	pF
90 to 170	$\frac{450}{f} + 1.2$
170+ to 250	$\frac{322}{f} + 1.2$

1/ The letter "f" represents specified frequency in kHz. A capacitance tolerance of ±5 % is permitted.

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NOTES:

(D) 1. The socket designed to mate with crystal holder HC-13/U is Part No. M12883/04-01. (See MIL-S-12883/4).
For Navy use, other sockets listed in MIL-STD-242 may be used.

(D) 2. NATO preferred type NXT13/A.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
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Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
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Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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